

HARDWARE

Team name:	Aurat
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Name of the Local Currency	Total Cost Local Currency	Total Cost (\$) U.S.A. Dollars
South Korea WON KRW	₩1.787.494,33	\$1.013,79

#	Component	Part name	Autor	Source	Quantity	Unit cost	Unit cost (US)	Total Cost	Total Cost
1	Shock Absorbers	ZERO mc 53mm spring shock absorber	ZERO mc	https://it	8	₩2.020,30	\$1,45	₩266.888,00	\$11,60
2	PLA	Sunlu pla+ 2.0	SUNLU	https:	1	₩33.361,00	\$21,79	₩33.361,00	\$21,79
3	TPU	Anycubic TPU 95a	Anycubic	https:	1	₩56.215,00	\$36,73	₩56.215,00	\$36,73
4	Linear Rails	Steel Bars (1mt)	Hardware Store	-----	10	₩316,71	\$0,21	₩3.167,10	\$2,07
5	Custom PCB	Custom PCB	Loris Berra and Futura Elettronica	https:	1	₩32.865,33	\$21,52	₩32.865,33	\$21,52
6	Microcontroller	Arduino Mega 2560 Pro	TECNOIOT	https:	1	₩42.410,33	\$27,77	₩42.410,33	\$27,77
7	Processor	Raspberry Pi 5	Raspberry Pi	https:	1	₩353.429,23	\$231,47	₩353.429,23	\$231,47
8	Raspberry Cooler	Raspberry Cooler	Raspberry pi	https:	1	₩12.370,32	\$8,10	₩12.370,32	\$8,10
9	Motor	Dc Motor With 90 Degree Reduction	School Supplies	-----	3	₩22.907,99	\$15,00	₩68.723,97	\$45,00
10	Motor with Encoder	Dc Motor With 90 Degree Reduction and Encoder	Tomvorks	https:	1	₩50.397,58	\$33,00	₩50.397,58	\$33,00
11	JST Connectors	JST Connectors Kit	School Supplies	-----	1	₩17.669,70	\$11,57	₩17.669,70	\$11,57
12	Microservo Motor	microservo motor hight torque mf90	Miuzei	https:	1	₩32.071,19	\$21,00	₩32.071,19	\$21,00
13	Cameras	OpenMV SingTown Cam H7 Plus Genuine-5MP High	OpenMV LLC	https:	2	₩218.389,53	\$143,00	₩436.779,06	\$286,00
14	ToF	vl6180x	Walfront	https:	6	₩20.036,86	\$13,12	₩120.221,16	\$78,72
15	Cables	Cables	Recycled from School Supplies	-----		₩0,00	\$0,00	₩0,00	\$0,00
16	Color Sensor	CQRobot TCS34725FN	QCRobot	https:	1	₩30.040,01	\$19,67	₩30.040,01	\$19,67
17	Capacitors	Capacitors	Recycled from School Supplies	-----		₩0,00	\$0,00	₩0,00	\$0,00
18	Resistors	Resistors	Recycled from School Supplies	-----		₩0,00	\$0,00	₩0,00	\$0,00
19	Voltage Regulators	7805 - lm317	Recycled from School Supplies	-----		₩0,00	\$0,00	₩0,00	\$0,00
20	Motor Driver	TB6612FNG	Shutao	https:	1	₩32.529,35	\$21,30	₩32.529,35	\$21,30
21	Led Strip	Led Strip	YIXISI	https:	1	₩22.861,00	\$15,00	₩22.861,00	\$15,00
22	Led Light Diffuser	Led Light Diffuser	Greenice	https:	1	₩1.582,00	\$8,00	₩1.582,00	\$8,00
23	Signaling Led Strip	Signaling Led Strip	Recycled	-----	1	₩0,00	\$0,00	₩0,00	\$0,00
24	Battery	11,1V 1600mAh 120C	Unnamed Brand	-----	2	₩61.457,00	\$40,00	₩122.914,00	\$80,00
25	Powerbank	Powerbank	Unnamed Brand	https:	1	₩24.620,00	\$16,17	₩24.620,00	\$16,17
26	Bumper Wheels	Bumper Wheels	Hardware store	-----	1	₩8.793,00	\$5,77	₩8.793,00	\$5,77
27	Big Bolts	Big Bolts	Hardware store	-----	1	₩8.793,00	\$5,77	₩8.793,00	\$5,77
28	Small Bolts	3mm	hardware store	-----	1	₩8.793,00	\$5,77	₩8.793,00	\$5,77
29	Handle	Handle	Recycled from School Supplies	-----	1	₩0,00	\$0,00	₩0,00	\$0,00

SOFTWARE

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Name of the Local Currency	Total Cost Local Currency	Total Cost (\$) U.S.A. Dollars
South Korea WON KRW	₩0,00	\$0,00

#	Name	Software's Tool/Library	Description	Source	Autor	Cost (Local)	Cost (USA)
1	Open CV	Library	Image processing library	OpenCV - Biblioteca abierta de Computer	Opencv.org	FREE	FREE
2	IDE Arduino	Software's Tool	IDE used for Arduino code	Software IDE Arduino	Arduino.cc	FREE	FREE
3	Numpy	Library	Array and math library	NumPy	Numpy developers	FREE	FREE
4	PySerial	Library	Serial communication library	pyserial.org/project/pyserial	Chris Liechti	FREE	FREE
5	gpiozero	Library	GPIO pin control library	gpiozero.readthedocs.io	Ben Nuttall	FREE	FREE
6	Adafruit TCS34725 (Python)	Library	Python color sensor driver	https://github.com/adafruit/Adafruit_CircuitPython_TCS34725	Adafruit Industries	FREE	FREE
7	Adafruit Blinka	Library	CircuitPython compatibility layer	pypi.org/project/Adafruit-Blinka	Adafruit Industries	FREE	FREE
8	OpenMV (sensor/image)	Library	MicroPython machine vision library	https://github.com/openmv/openmv	OpenMV LLC	FREE	FREE
9	Wire.h	Library	I2C communication library	github.com/arduino/ArduinoCore-avr	Arduino.cc	FREE	FREE
10	Adafruit Unified Sensor	Library	Unified sensor abstraction layer	github.com/adafruit/Adafruit_Sensor	Adafruit Industries	FREE	FREE
11	Adafruit BNO055	Library	IMU 9-axis sensor library	github.com/adafruit/Adafruit_BNO055	Adafruit Industries	FREE	FREE
12	Adafruit VL6180X	Library	ToF distance sensor library	github.com/adafruit/Adafruit_VL6180X	Adafruit Industries	FREE	FREE
13	Servo.h	Library	Servo motor control library	github.com/arduino-libraries/Servo	Michael Margolis	FREE	FREE
14	Python Std Libraries	Library	Core system and threading utilities	github.com/python/cpython	Python Software Foundation	FREE	FREE
15	avr/wdt.h	Library	Watchdog timer control library	github.com/avrdudes/avr-libc	AVR-LibC Team	FREE	FREE
16	Visual Studio Code	Software's Tool	IDE used for Python code	github.com/microsoft/vscode	Microsoft	FREE	FREE
17	Blender	Software's Tool	3D creation suite	github.com/blender/blender	Blender Foundation	FREE	FREE
18	EasyEDA	Software's Tool	PCB design tool	easyeda.com	JLCPCB	FREE	FREE
19	OpenMV IDE	Software's Tool	Machine vision development IDE	github.com/openmv/openmv-ide	OpenMV LLC	FREE	FREE
20	C/C++ Standard Library (math.h)	Library	Standard math library	github.com/gcc-mirror/gcc	GCC Developers	FREE	FREE
21	utility/imumaths.h	Library	Vector and matrix math library	github.com/adafruit/Adafruit_BNO055	Adafruit Industries	FREE	FREE
22	Mahotas	Library	Computer vision and image processing	github.com/luispedro/mahotas	Luis Pedro Coelho	FREE	FREE
23	Scikit-learn (sklearn)	Library	Machine learning library	github.com/scikit-learn/scikit-learn	Scikit-learn Developers	FREE	FREE
24	Matplotlib	Library	Data visualization library	github.com/matplotlib/matplotlib	Matplotlib Developers	FREE	FREE
25	Seaborn	Library	Statistical data visualization library	github.com/mwaskom/seaborn	Michael Waskom	FREE	FREE